- Finding the expansion of e^x about $x = 0$

$f(x) = e^x \Rightarrow f(0) = e^0 = 1$

$f'(x) = e^x \Rightarrow f'(0) = e^0 = 1; f''(0) = f'''(0) = f''''(0) = \ldots\ldots = 1$

$f(x) = e^x = 1 + (x-0)\frac{1}{1!} + (x-0)^2.\frac{1}{2!} + (x-0)^3.\frac{1}{3!} + \ldots\ldots$

$\Rightarrow \quad e^x = 1 + \frac{x}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \ldots\ldots$

1.6 Integral Calculus

If $F(x)$ is anti-derivative of $f(x)$. That is, continuous and differentiable in (a, b), then we write $\int_{x=a}^{x=b} f(x)\,dx = F(b) - F(a)$. Here $f(x)$ is integrand

If $f(x) > 0\ \forall a \le x \le b, then\ \int_a^b f(x)\,dx$ represents the shaded area in the given figure.

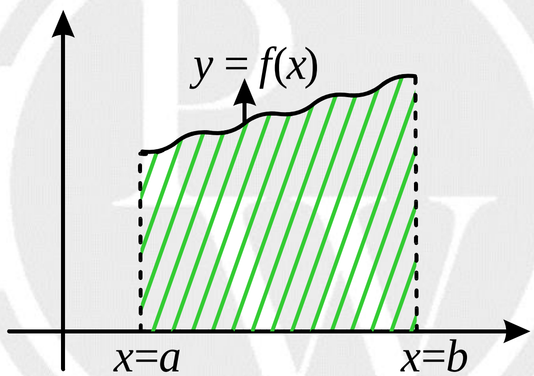

Fig.1. 6. Integration of continuous function

1.6.1 Mean Value Theorem of Integration

If $f(x)$ is continuous in $[a, b]$ and differentiable in (a, b) then '∃' atleast one-point $c \in (a, b)$ such that

$f(c) = \frac{\int_a^b f(x)dx}{(b-a)}$

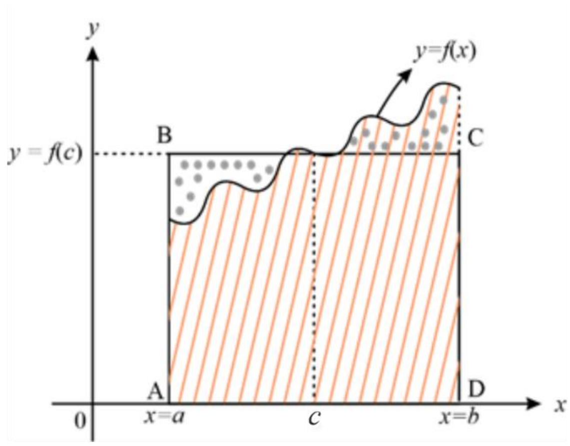

Fig. 1.7. Mean value of integration

1.7. Newton-Leibnitz Rule

If $f(x)$ is continuously differentiable and $\phi(x)$, $\Psi(x)$ are two functions for which the 1st derivative exists, then

$$\frac{d}{dx}\left(\int_{\phi(x)}^{\psi(x)} f(x)dx\right) = f\big(\psi(x)\big).\psi'(x) - f\big(\phi(x)\big).\phi'(x)$$

Example: $\frac{d}{dx}\left(\int_{x}^{x^2} \sin x\, dx\right) = \sin(x^2)\,.\,2x - \sin x\,.\,1 = 2x\sin(x^2) - \sin x$

1.8. Some Standard Integrals

1. $\int x^n dx = \frac{x^{n+1}}{n+1} + C,\ (n \neq -1)$

2. $\int \frac{1}{x} dx = \log_e |x| + C$

3. $\int \sin x\, dx = -\cos x + C$

4. $\int \cos x\, dx = \sin x + C$

5. $\int \frac{f'(x)}{f(x)} dx = \log_e |f(x)| + C$

6. $\int \tan x\, dx = -\int -\frac{\sin x}{\cos x} dx = -\log_e |\cos x| + C$

$\Rightarrow \int \tan x\, dx = \log_e |\sec x| + C$

7. $\int \cot x\, dx = \int \frac{\cos x}{\sin x} dx = \log_e |\sin x| + C = -\log_e |\operatorname{cosec} x| + C$

8. $\int \sec x\, dx = \int \frac{\sec x(\sec x + \tan x)}{(\sec x + \tan x)} dx = \log_e |\sec x + \tan x| + C$

9. $\int \operatorname{cosec} x dx = \log_e |\operatorname{cosec} x - \cot x| + C$

10. $\int a^x dx = \frac{a^x}{\log_e a} + C$

11. $\int \frac{1}{x.\log_e a} dx = \log_a x + C$

12. $\int x^x (1 + \log_e x)\, dx = x^x + C$

13. $\int f(x)\,.\,f'(x) dx = \frac{1}{2}\big(f(x)\big)^2 + C$

14. $\int \frac{f'(x)}{\sqrt{f(x)}} dx = 2.\sqrt{f(x)} + C$

15. If $f(x)$, $g(x)$ are two functions. that are differentiable, then

$\int f(x)\, g(x) dx = f(x).\int g(x)\, dx - \int [f'(x)\, g(x)] dx + C$

Before integrating the product, the functions $f(x)$ and $g(x)$ are to be arranged according to the ILATE Principle.

Here, ILATE stands for INVERSE LOGARITHMIC ALGEBRAIC TRIGONOMETRIC EXPONENTIAL.

1.9 Properties of Definite Integrals

1. If $f(x)$ is differentiable in interval (a, b), then $\int_a^b f(x)\,dx = -\int_b^a f(x)\,dx$

2. If $\exists$ a point $c \in (a, b)$ such that $f(x)$ is not differentiable, then

$\int_a^b f(x)\,dx = \int_a^c f(x)\,dx + \int_c^b f(x)\,dx$

3. If $f(x)$ is continuously differentiable function,

$\int_{-a}^a f(x)\,dx = 2\times\int_0^a f(x)\,dx$; if $f(-x) = f(x)$, ("$f(x)$ is even function")

$= 0$; if $f(-x) = -f(x)$, ("$f(x)$ is odd function")

4. $\int_0^{2a} f(x)\,dx = 2\times\int_0^a f(x)\,dx$, if $f(2a-x) = f(x)$

5. $\int_a^b f(x)\,dx = \int_a^b f(a+b-x)\,dx$

6. $\int_a^b \frac{f(x)}{f(x)+f(a+b-x)} = \left(\frac{b-a}{2}\right)$

Example:

(i) $\int_0^{\pi/2} \frac{\sin x}{\sin x+\cos x} = \frac{\pi}{4}$

(ii) $\int_0^{\pi/2} \frac{1}{1+\sqrt{\tan x}}\,dx = \int_0^{\pi/2} \frac{1}{1+\left(\frac{\sqrt{\sin x}}{\sqrt{\cos x}}\right)}\,dx = \int_0^{\pi/2} \frac{\sqrt{\cos x}}{\sqrt{\cos x}+\sqrt{\sin x}}\,dx = \frac{\pi}{4}$

(iii) $\int_2^3 \frac{\sqrt{x}}{\sqrt{x}+\sqrt{5-x}} = \left(\frac{3-2}{2}\right) = \frac{1}{2}$

(iv) $\int_0^{\pi/2} \frac{\sqrt{\tan x}}{\sqrt{\tan x}+\sqrt{\cot x}}\,dx = \frac{\pi}{4}$

7. $\int_0^{\pi/2} \sin^m x\,dx = \int_0^{\pi/2} \cos^m x\;dx = \frac{(m-1)\times(m-3)\times(m-5)}{m\times(m-2)\times(m-4)}\times\ldots\left(\frac{1}{2}\right)$ (or) $\frac{2}{3}\times K$

Where $K = \pi/2$ if m is even

$= 1$ if m is odd.

8. $\int_0^{\pi} \frac{dx}{a^2\cos^2 x+b^2\sin^2 x} = \frac{\pi}{ab}$

9. $\int_0^{\pi/2} \frac{dx}{a^2\cos^2 x+b^2\sin^2 x} = \frac{\pi}{2ab}$

1.10 Length of a Curve

(a) The length of the arc of the curve $y = f(x)$ between the points where x = a and x = b is $s = \int_a^b \sqrt{\left[1+\left(\frac{dy}{dx}\right)^2\right]}dx$

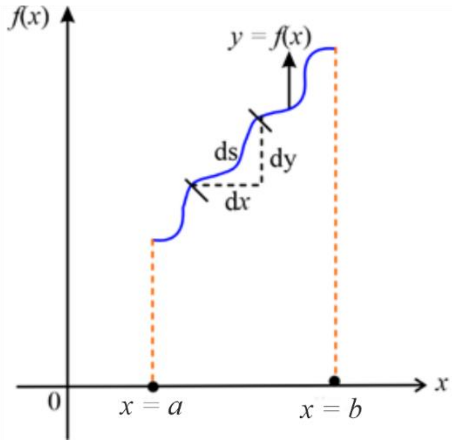

Fig.1.8. Length of the curve

(b) The length of the arc of the curve $x = f(y)$ between the points where y = a and y = b, is $s = \int_a^b \sqrt{\left[1 + \left(\frac{dx}{dy}\right)^2\right]} dy$

(c) The length of the arc of the curve $x = f(t), y = f(t)$ between the points where t = a and t = b, is $s = \int_a^b \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} dt$

(d) The length of the arc of the curve $r = f(\theta),$ between the points where $\theta = \alpha$ and $\theta = \beta$, is $s = \int_\alpha^\beta \sqrt{\left[r^2 + \left(\frac{dr}{d\theta}\right)^2\right]} d\theta$

1.11 Surface Area of Solid generated by revolving a curve about a fixed axis

Elemental Surface Area

$dA = 2\pi y \times ds = 2\pi y ds$

$\Rightarrow$ Total surface area = A = $\int_{x=a}^{x=b} 2\pi y \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx$

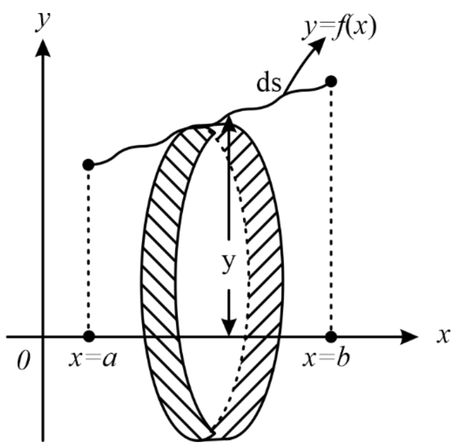

Fig.1.9. Surface area

1.12 Volume of the solid

A. The volume of the solid obtained by revolving the curve y = $f(x)$ between the lines $x = a$ and $x = b$ is given by

$\Rightarrow \quad V \approx \int_{x=a}^{x=b} \pi y^2 dx$

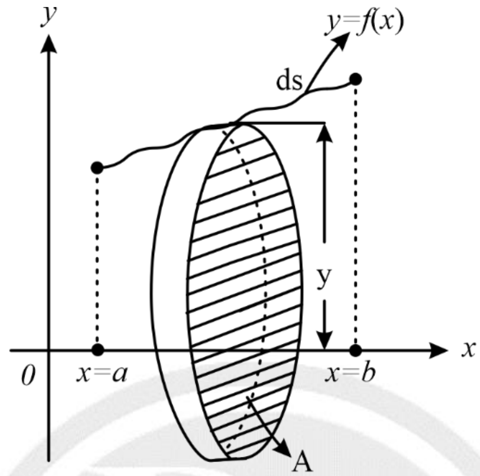

Fig. 1.10. Volume of the solid

B. **Revolution about the y-axis.** Interchanging x and y in the above formula, we see that the volume of the solid generated by the revolution, about y-axis, of the area, bounded by the curve $x = f(y)$, the y-axis and the abscissa y = a, y = b is $\int_a^b \pi x^2 dy$.

1.13 Gamma Function

The integral $\int_0^\infty e^{-x}.x^{n-1}\,dx, (n > 0)$ is called Gamma function of n. It is denoted by $\Gamma n = \int_0^\infty e^{-x} x^{n-1} dx$.

Note : $$\int_0^{\pi/2} \sin^m x \cos^n x dx = \frac{\Gamma\left(\frac{m+1}{2}\right)\Gamma\left(\frac{n+1}{2}\right)}{2\Gamma\left[\frac{m+n+2}{2}\right]}$$

Where Γ(x) is called the gamma function.

1.13.1 Properties of Gamma Function

(i) $\Gamma n = (n-1)!$

(ii) $\Gamma(n+1) = (n)!$

(iii) $\Gamma(n+1) = n\Gamma n$

(iv) $\Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}$

1.14 Beta Function

The function β $(m, n) = \int_0^1 x^{m-1}.(1-x)^{n-1}\,dx$ $(m, n > 0)$ is called β function of m and n.